

13. 设平面区域 D 由曲线 $y = \sqrt{3(1-x^2)}$ 与直线 $y = \sqrt{3}x$ 及 y 轴所围成. 计算二重积分

$$\iint_D (x^2 + y^2) dx dy.$$

$$x^2 + \frac{y^2}{3} = 1$$

$$r^2 \cos^2 \theta + \frac{r^2}{3} \sin^2 \theta = 1$$

$$r = \frac{1}{\cos^2 \theta + \frac{\sin^2 \theta}{3}}$$

$$\int_0^{\frac{1}{\sqrt{2}}} dx \int_{\sqrt{3}x}^{\sqrt{3(1-x^2)}} (x^2 + y^2) dy$$

$$= \int_0^{\frac{1}{\sqrt{2}}} x^2 \sqrt{3(1-x^2)} dx + \frac{1}{3} \int_0^{\frac{1}{\sqrt{2}}} [3(1-x^2)]^{\frac{3}{2}} dx$$

$$- 2\sqrt{3} \int_0^{\frac{1}{\sqrt{2}}} x^3 dx$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos^2 \theta d\theta + \sqrt{3} \int_0^{\frac{\pi}{4}} \cos^4 \theta d\theta$$

$$- 2\sqrt{3} \frac{x^4}{4} \Big|_0^{\frac{1}{\sqrt{2}}}$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos^2 \theta d\theta + \sqrt{3} \int_0^{\frac{\pi}{4}} \cos^2 \theta - \cos^2 \theta d\theta$$

$$- 2\sqrt{3} \cdot \frac{1}{16}$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{2} \int_0^{\frac{\pi}{4}} (1 + \cos 2\theta) d\theta - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{2} \left[\frac{\pi}{4} + \frac{1}{2} \cdot 1 - 0 \right] - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{8} \pi + \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{8} (\pi + 1)$$

设 $x = r \cos \theta$

$y = \sqrt{3} r \sin \theta$

$$\text{面积} = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\theta \int_0^{\frac{1}{\cos^2 \theta + \frac{\sin^2 \theta}{3}}} r^2 \left| \frac{\partial x}{\partial r} \frac{\partial y}{\partial \theta} - \frac{\partial x}{\partial \theta} \frac{\partial y}{\partial r} \right| dr$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\theta \int_0^{\frac{1}{\cos^2 \theta + \frac{\sin^2 \theta}{3}}} r^2 \cdot \sqrt{3} r dr$$

$$= \sqrt{3} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(\cos^2 \theta + \frac{\sin^2 \theta}{3})^2}{4} d\theta$$

$$= \frac{\sqrt{3}}{4}$$

13. 设平面区域 D 由曲线 $y = \sqrt{3(1-x^2)}$ 与直线 $y = \sqrt{3}x$ 及 y 轴所围成. 计算二重积分

$$\iint_D (x^2 + y^2) dx dy.$$

$$x^2 + \frac{y^2}{3} = 1$$

$$\int_0^{\frac{1}{\sqrt{3}}} dx \int_{-\sqrt{3(1-x^2)}}^{\sqrt{3(1-x^2)}} x^2 + y^2 dy$$

一、思路.

① 画 D

② $\iint_D (x^2 + y^2) dx dy$

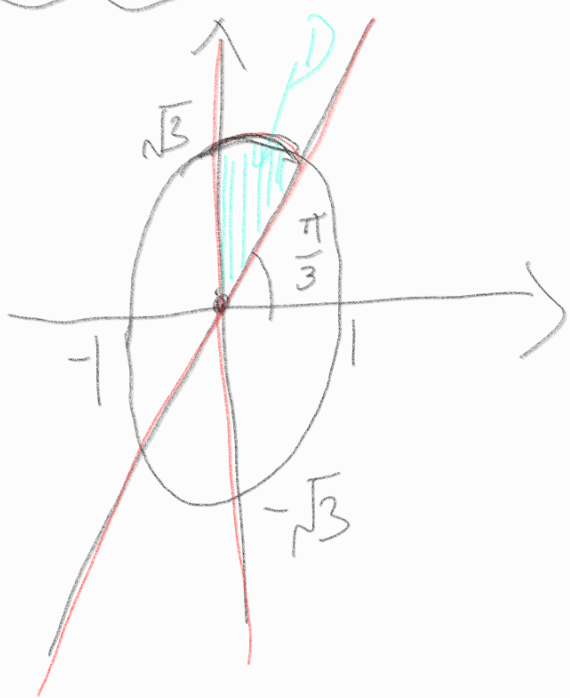
→ 极坐标 ✓

→ D 关于 $y=x$ 对称 ✓

$$\frac{1}{2} \iint_D (x^2 + y^2) dx dy$$

→ xy 坐标
硬计算

$y = \sqrt{3(1-x^2)}$, y 轴, $y = \sqrt{3}x \rightarrow$ 极坐标



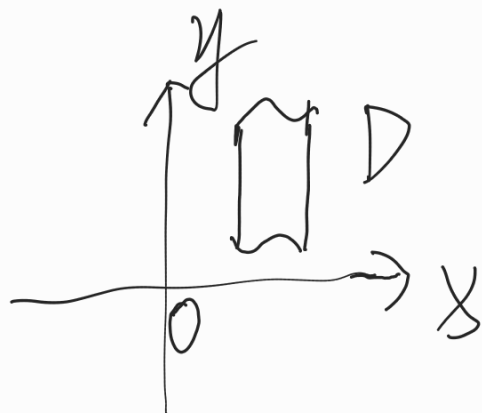
$$\frac{y}{x} = \sqrt{3} = \tan \theta$$

$$\theta = \frac{\pi}{3}$$

$$y^2 = 3(1-x^2)$$

$$\frac{x^2}{1^2} + \frac{y^2}{(\sqrt{3})^2} = 1$$

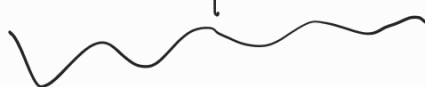
二重积分 $\longrightarrow$ 累次积分



$$\int_0^C \int_0^x dy dx$$



$$\int_0^x \int_0^y dx dy$$



$$\iint_D x^2 + y^2 \, dx \, dy$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$y = \sqrt{3(1-x^2)}$$

$$r^2 \sin^2 \theta = 3 - 3r^2 \cos^2 \theta$$

$$r = \frac{3}{3\cos^2 \theta + \sin^2 \theta}$$

$$\textcircled{1} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\theta \int_0^{\frac{3}{3\cos^2 \theta + \sin^2 \theta}} r^2 \cdot \underline{r \, dr}$$



$$u = x = r \cos \theta$$

$$v = \sqrt{3}y = \sqrt{3} r \sin \theta$$

$$y = \sqrt{3(1-x^2)} \Rightarrow \frac{x^2}{1^2} + \frac{y^2}{3} = 1$$

$$u^2 + v^2 = 1 \Rightarrow \frac{x^2}{1^2} + \frac{y^2}{3} = 1$$

$$\iint x^2 + y^2 \, dx \, dy$$

$$x = r \cos \theta$$

$$y = \sqrt{3} r \sin \theta$$

$$f(x, y) \Rightarrow f(r \cos \theta, \sqrt{3} r \sin \theta)$$

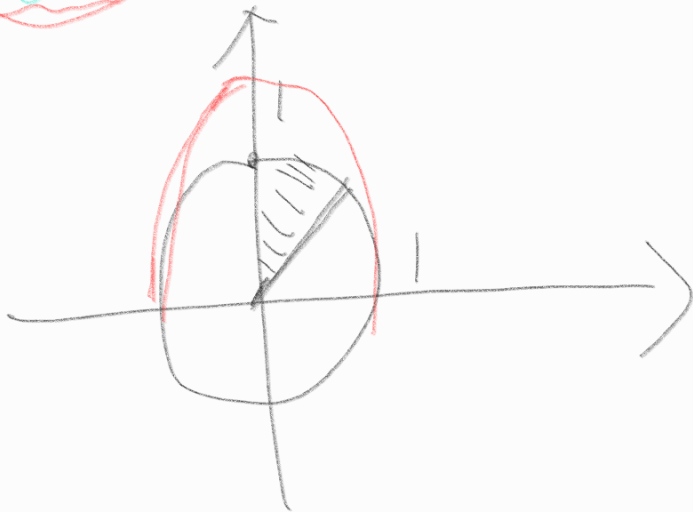
$$D_{xy} \Rightarrow D_{r, \theta}$$

$$|J| = \sqrt{3} r$$

$$dG_{xy} \Rightarrow$$

$$\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} dG_{r, \theta}$$

$$\sqrt{3}$$



$$S_{\text{sector}} = \pi ab = \pi \cdot 1 \cdot \sqrt{3}$$

$$S_{\text{circle}} = \pi r^2 = \pi$$

$$D_{xy} \left\{ \begin{aligned} y &= \sqrt{3(1-x^2)} \\ \text{y轴}, x=0 &\Rightarrow \\ y &= \sqrt{3}x \end{aligned} \right.$$



$$\begin{aligned} \sqrt{3} r^2 \sin^2 \theta &= \sqrt{3} (1 - r^2 \cos^2 \theta) \\ r^2 \sin^2 \theta &= 1 - r^2 \cos^2 \theta \\ r^2 &= 1 \\ r &= 1 \\ r \cos \theta &= 0 \\ \theta &= \frac{\pi}{2} \end{aligned}$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta$$

$$\begin{aligned} \sqrt{3} r \sin \theta &= \sqrt{3} r \cos \theta \\ \theta &= \frac{\pi}{4} \end{aligned}$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta$$

$$\int_0^1$$

$$\frac{r^2 \cos^2 \theta + 3r^2 \sin^2 \theta}{|J|} dr$$

② 广义极坐标 (按元 ||J||)

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^1 r^2 + 2r^2 \sin^2 \theta \left| \begin{array}{cc} \frac{\partial x}{\partial r} = \cos \theta & \frac{\partial x}{\partial \theta} = r \sin \theta \\ \frac{\partial y}{\partial r} = \sqrt{3} \sin \theta & \frac{\partial y}{\partial \theta} = \sqrt{3} r \cos \theta \end{array} \right| dr$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^1 (r^2 + 2r^2 \sin^2 \theta) \cdot \sqrt{3} r dr$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \cdot \int_0^1 \sqrt{3} r^3 dr + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^1 2\sqrt{3} r^3 \sin^2 \theta dr$$

$$= \frac{\pi}{4} \cdot \frac{\sqrt{3}}{4} (1-0) + 2\sqrt{3} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 \theta \cdot \frac{1}{4} d\theta$$

$$= \frac{\sqrt{3}}{16} \pi + \frac{\sqrt{3}}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 \theta d\theta \quad \left| \int \sin^2 \theta d\theta \right.$$

$$= \frac{\sqrt{3}}{16} \pi + \frac{\sqrt{3}}{2} \cdot \frac{1}{2} \left(\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos 2\theta d\theta \right) = \int \frac{1 - \cos 2\theta}{2} d\theta$$

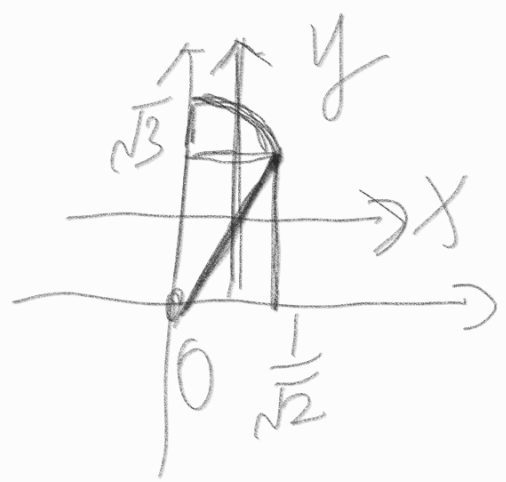
$$= \frac{\sqrt{3}}{16} \pi + \frac{\sqrt{3}}{4} \cdot \left(\frac{\pi}{4} - \frac{1}{2} \sin 2\theta \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \right) = \frac{1}{2} \left(\int d\theta - \int \cos 2\theta d\theta \right)$$

$$= \frac{\sqrt{3}}{16} \pi + \frac{\sqrt{3}}{16} \pi - \frac{\sqrt{3}}{8} (0-1)$$

$$= \frac{\sqrt{3}}{8} (\pi + 1)$$

(3)

$$\iint_D x^2 + y^2 dx dy$$



$$\int_0^{\frac{1}{\sqrt{2}}} dx \int_{\sqrt{3}x}^{\sqrt{3(1-x^2)}} x^2 + y^2 dy$$

$$y = \sqrt{3(1-x^2)}$$

$$y = \sqrt{3}x$$

$$1-x^2 = x^2$$

$$= \int_0^{\frac{1}{\sqrt{2}}} \left(x^2 y + \frac{y^3}{3} \right) \bigg|_{\sqrt{3}x}^{\sqrt{3(1-x^2)}} dx$$

$$= \int_0^{\frac{1}{\sqrt{2}}} x^2 \sqrt{3(1-x^2)} + \sqrt{3} (1-x^2)^{\frac{3}{2}} - 2\sqrt{3} x^3 dx$$

$$= \sqrt{3} \int_0^{\frac{1}{\sqrt{2}}} x^2 \sqrt{1-x^2} dx + \sqrt{3} \int_0^{\frac{1}{\sqrt{2}}} (1-x^2)^{\frac{3}{2}} dx - 2\sqrt{3} \int_0^{\frac{1}{\sqrt{2}}} x^3 dx$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos \theta d\sin \theta + \sqrt{3} \int_0^{\frac{\pi}{4}} |\cos \theta| \cdot \cos^2 \theta d\sin \theta - 2\sqrt{3} \cdot \frac{(\frac{1}{\sqrt{2}})^4 - 0}{4}$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos^2 \theta d\theta + \sqrt{3} \int_0^{\frac{\pi}{4}} \cos^4 \theta d\theta - \frac{\sqrt{3}}{8}$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos^2 \theta d\theta + \sqrt{3} \int_0^{\frac{\pi}{4}} \cos^2 \theta (1 - \sin^2 \theta) d\theta - \frac{\sqrt{3}}{8}$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta - \frac{\sqrt{3}}{8}$$

$$= \sqrt{3} \int_0^{\frac{\pi}{4}} \frac{\cos 2\theta + 1}{2} d\theta - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{4} \sin 2\theta \Big|_0^{\frac{\pi}{4}} + \frac{\sqrt{3}}{2} \times \frac{\pi}{4} - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{4} \times (1-0) + \frac{\sqrt{3}}{8} \pi - \frac{\sqrt{3}}{8}$$

$$= \frac{\sqrt{3}}{8} (\pi + 1)$$

